



A perspective on quantum Fintech

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Abstract

Recently, FinTech has emerged as a hot topic, utilizing technology to enhance and innovate financial services and products. This expanding topic requires theoretical models and innovative methodologies to forward new business challenges. In today's world, where vast amounts of data are generated every day, the need for computers capable of accurate predictive computations is becoming increasingly critical. Consequently, many financial institutions are turning to quantum computing, which promises to analyze large datasets and deliver results more quickly and accurately than any classical computer ever could. On the other hand, knowledge about quantum computing is not yet widely diffused in finance communities. In this work, after providing a gentle introduction to quantum mechanics, we review the state of the art of quantum computing in Fintech, touching such themes as stochastic modeling, optimization, and machine learning. Theoretical results, as well as practical solution, are discussed with the associated challenges.

Keywords Economics · Actuarial · Financial sciences · Quantum computing · Fintech

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1 Introduction

Quantum algorithms have the potential to revolutionize various financial problems by providing efficient solutions for important techniques such as stochastic modelling, optimization, and machine learning. These methods can be applied to problems such as option pricing, risk management, and portfolio optimization, which are crucial for financial institutions. Quantum algorithms can also be used to model complex financial systems, predict market trends, and optimize investment portfolios. The applicability of quantum algorithms to financial problems is vast, and researchers are actively exploring new ways to apply these algorithms to real-world financial problems. In addition, novel ideas emerge from the application of the fundamental ideas of quantum computing to apparently completely unrelated situations, making also the theoretical landscape benefit. Finally, the potential weaknesses of cryptographic algorithms has given rise to a new branch of cryptography, post/quantum cryptography (Bernstein and Lange 2017). Within their analysis of the risks and benefits of quantum computing, Scholten et al. (2024) remark, however, that it can be expected that the computations which quantum computers will be economically-impactful instead of cryptographically-relevant.

Quantum Monte Carlo integration and gradient estimation are two quantum algorithms that have the potential to provide quadratic speedup over classical methods. These algorithms can be used to estimate complex integrals and gradients, which are essential for various financial problems such as option pricing and risk management. However, to achieve early fault-tolerant feasibility and actual speedup, more work is required to reduce the amount of quantum resources required for these algorithms. This includes developing more efficient quantum circuits, reducing the number of qubits required, and improving the accuracy of the algorithms.

Financial optimization problems can be classified into several categories, including continuous (convex or non-convex), discrete, or mixed. Quantum algorithms can be applied to these problems to provide efficient solutions. For example, quantum algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE) can be used to solve continuous optimization problems, while quantum algorithms such as the Quantum Alternating Projection Algorithm (QAPA) can be used to solve discrete optimization problems. Mixed optimization problems can also be solved using quantum algorithms such as the Quantum Mixed-Integer Linear Programming (QMILP) algorithm.

Quantum machine learning has the potential to revolutionize various classical problems in finance, including portfolio optimization, risk management, and option pricing. The advantages of quantum machine learning include the ability to provide efficient solutions for complex problems, improve the accuracy of predictions, and reduce the risk of errors. However, there are also challenges associated with quantum machine learning, including the requirement for large amounts of quantum resources, the need for efficient quantum algorithms, and the challenge of interpreting the results of quantum machine learning models. Despite these challenges, researchers are actively exploring the potential of quantum machine learning for financial problems.

For a general introduction to quantum computing, we recommend the book by Nielsen and Chuang (2010). Numerous works have reviewed the application of quan-

tum computing to finance. Written with an eye to a readership of physicists, the survey by Herman et al. (2023) focused on how quantum computing can be applied to solve financial problems more efficiently, including derivative pricing, risk modeling, portfolio optimization, and fraud detection. The state-of-the-art, prospects, and potential benefits of quantum computing for finance, including specific applications in simulation, optimization, and machine learning are outlined in Egger et al. (2020), including demonstrations of some quantum algorithms on IBM Quantum systems. Lopatnikova et al. (2021) aimed at providing statisticians and data scientists with the language and the basic knowledge needed to interact successfully with quantum algorithms designers. Moreover, there are some interesting references (Di Nola et al. 2019, 2021) to the theoretical foundations of quantum approach and the relations with the domain of suitable logical and mathematical structures.

This work:

- (i) aims at providing a unique perspective to researchers in the field of Mathematical Methods of Economics, Actuarial and Financial Sciences;
- (ii) focuses on newer references (mostly from 2022 onwards) in a field which is evolving at a very fast pace. As a methodological note, we have preferred to include such recent works, since the seminal works that shaped a specific topic are referenced therein and the interested reader may find them.

2 Quantum computing

Quantum mechanics is the fundamental framework that explains a wide range of physical phenomena, particularly at the atomic scale, where classical mechanics fails. It provides the mathematical and conceptual tools to understand and predict the behavior of particles like electrons, photons, and atoms. In quantum mechanics, the state of a physical system is described by a vector in a finite-dimensional or infinite-dimensional Hilbert space. This space is a complete inner-product space that provides the necessary structure to accommodate the principles of quantum mechanics, such as superposition and entanglement. Each possible state of the system corresponds to a unique vector in this space, and observable quantities are represented by operators acting on these vectors.

Quantum computing is a computing paradigm based on the use of the tools of quantum mechanics to perform, and describe, computation. Notably, quantum computing enjoys the mathematical simplicity of considering finite-dimensional Hilbert spaces.

Qubits

The fundamental unit of quantum information is the *quantum bit*, or *qubit*. Unlike a classical bit, which can only be in one of two definite states, 0 or 1, a qubit can exist in a superposition of both states simultaneously. Reflecting the superposition principle of dynamical states, a characteristic of any wave in general, the state of a qubit can be expressed as a superposition of the states in a basis of the bidimensional vector space where qubits lie. Such superposition is fundamentally different from a probabilistic mixture of states. In standard Dirac notation, a state vector ψ is represented by the “ket” $|\psi\rangle$, while its conjugate transpose (adjoint) will be written as the “bra” $\langle\psi| = |\psi\rangle^\dagger$.

The inner product of $|\phi\rangle$ and $|\psi\rangle$ is $\langle\phi|\psi\rangle$. In particular, the set $\{|0\rangle, |1\rangle\}$ forms an orthonormal basis, known as the computational basis, on the space of qubits, implying that any state vector $|\psi\rangle$ can be written as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (1)$$

where the coefficients $\alpha, \beta \in \mathbb{C}$ are known as amplitudes and contain information about the likelihood of the qubit being found in either the state $|0\rangle$ or $|1\rangle$ upon measurement. An essential aspect is that the actual probabilities of measuring the qubit in the state $|0\rangle$ and $|1\rangle$, respectively, are proportional to $|\alpha|^2 = \alpha^* \alpha$ and $|\beta|^2$ (the asterisk denotes complex conjugation). The normalization condition $\langle\psi|\psi\rangle = 1$ is convenient for the probabilistic interpretation of superposition states in that it ensures that the total probability of all possible outcomes in the computational basis (measuring $|0\rangle$ or $|1\rangle$) is equal to 1.

Quantum gates

Operations on qubits are performed using quantum gates, which are unitary operators acting on the state space. Gates manipulate qubit states by altering the amplitudes α and β , thus transforming the state vector $|\psi\rangle$. Due to the linearity of quantum mechanics, any physically valid quantum operation must preserve the norm of the state vector, ensuring that the total probability remains constant. As a result, the quantum evolution of a closed system in discrete time is inherently unitary and reversible. The state of the system can be perfectly recovered at any point in time, unless a measurement is made, which introduces non-unitary and irreversible dynamics. When a qubit in an arbitrary state $|\psi\rangle$ is measured in a basis, the superposition state $|\psi\rangle$ *collapses* to one of the basis states.

After measurement, the qubit is found in the state corresponding to the observed outcome, and any prior superposition is lost. In a sense, measurement does not record a property, it creates one. This probabilistic nature of quantum measurement is a distinctive feature of quantum mechanics, differing fundamentally from the deterministic predictions of classical mechanics.

The outcome of a projective measurement is one of the eigenstates of the operator associated to the measurements. Repeating the same measurement would then yield the same outcome. However, the probabilities for a measurement in a different basis would be unaffected. The order in which two distinct measurements are performed, in general, matters. Only if two operators commute the joint measurements is meaningful. The probabilities of outcomes are the squares of the magnitudes of the amplitudes of the state written as a superposition of the eigenvectors of the operators on which measurement is made. Measuring $|\psi\rangle$ as in (1) yields $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$. The *expected value* of the measurement with respect to an Hermitian operator M is equal to $\langle\psi|M|\psi\rangle$.

The evolution over time of a closed quantum system is described by an operator U , which must be unitary ($U^{-1} = U^\dagger$). In continuous time, the dynamics of a closed quantum system are governed by the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad (2)$$

where $\hbar$ is the reduced Planck constant (experimentally determined to be about $1.05457 \cdot 10^{-34}$ kg m²/s) and H is the Hamiltonian of the system, which must be an Hermitian operator to preserve the norm of $|\psi(t)\rangle$ in the course of time. Hence, H is diagonalizable and has real eigenvalues. For any two given times t_1 and t_2 , for the linear superposition of states to be preserved in the course of time, the correspondence between the state of the system at time t_1 , $|\psi(t_1)\rangle$, and $|\psi(t_2)\rangle$ must be given by a linear operator (the *time development* operator)

$$|\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle.$$

When H is Hermitian, each $U(t_2, t_1)$ in the family of operators parameterized by t_1 and t_2 must be unitary. Conversely, any unitary operator U can be written as $\exp(iK)$ for some Hermitian operator K . There is thus a one-to-one correspondence between the continuous time description using Hamiltonians and the discrete-time description of dynamics using unitary operators.

Note that the evolution of a quantum state depends on the amplitudes and not on the probabilities of measurement outcomes, producing effects such as quantum interference. Differently from probabilities, amplitudes can take any complex value, including negative ones, and this can make probabilities cancel out when considering the several different ways (histories) a single configuration can occur in the evolution of a system from a fixed initial state.

The Pauli matrices are a set of three 2×2 complex matrices that are of fundamental importance in quantum mechanics, particularly in the study of spin- $\frac{1}{2}$ particles and qubits. These matrices are typically denoted by σ_x , σ_y , and σ_z , and are defined as follows:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each of these matrices is Hermitian and unitary.

The Pauli matrices, along with the 2×2 identity matrix I , form the generators of the Pauli group. The Pauli group consists of matrices of the form $\{I, \sigma_x, \sigma_y, \sigma_z\}$, each multiplied by a phase factor of ± 1 or $\pm i$. The Pauli group is an important tool in quantum information theory, particularly in the context of error correction and the study of entanglement.

One of the key properties of the Pauli matrices is that they satisfy the following commutation and anticommutation relations:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k, \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij}I,$$

where $[A, B] = AB - BA$ is the commutator, $\{A, B\} = AB + BA$ is the anticommutator, ϵ_{ijk} is the Levi-Civita symbol, and δ_{ij} is the Kronecker delta. Pauli matrices generate rotations in the Bloch sphere, which represents the state space of a qubit. In quantum mechanics, the Pauli matrices are used to represent the spin operators for spin- $\frac{1}{2}$ particles. For example, the spin operator in the x -direction is given by $\hat{S}_x = \frac{\hbar}{2}\sigma_x$. The eigenvalues of the Pauli matrices, ± 1 , correspond to the possible measurement

outcomes of spin in the respective directions. The Pauli matrices, together with the identity, provide a basis for the 2×2 Hermitian matrices. The Pauli matrices are used extensively in quantum computing to describe quantum gates, denoted X, Y, and Z, which correspond to the application of σ_x , σ_y , and σ_z on a qubit.

Observables

There exist two categories of quantum operators: unitary transformations and observables. Unitary operators are isometries (they preserve the norm) and are used for change of basis. Observables are Hermitian (self-adjoint) operators. They are not isometries but their eigenvalues are real and their eigenvectors span the entire space. The possible outcomes of a measuring an observable correspond to its eigenvalues. Formally, quantum measurement is a collection of operators $\{M_m\}$ that correspond to outcomes m . When such measurement is applied to a quantum state $|\psi\rangle$, the probability of obtaining outcome m is given by $p_m = \langle\psi|M_m^\dagger M_m|\psi\rangle$. After the measurement, the state of the system is $\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$. The probabilities of all possible outcomes must sum to unity, i.e., $\sum_m p_m = 1$, for any quantum state. This requirement leads to the completion relation $\sum_m M_m^\dagger M_m = I$, which must be satisfied by the measurement operators.

A projection operator P is idempotent, i.e. such that $P^2 = P$. Given a ket ϕ , the operator $|\phi\rangle\langle\phi|$ is Hermitian and projects onto the subspace of all scalar multiples of $|\phi\rangle$. For example, the projection measurement operator that projects onto the state $|0\rangle$ is given by $P_0 = |0\rangle\langle 0|$. Restricting the operators in the collection to be *projection operators*, measurement can be described by the spectral decomposition of an operator with respect to some orthonormal basis. The set of all projection operators corresponding to a complete orthonormal basis satisfies the *completeness relation*, often called decomposition of identity:

$$\sum_i |\phi_i\rangle\langle\phi_i| = I,$$

where $\{|\phi_i\rangle\}$ represents the complete set of orthonormal basis states, and I is the identity operator in the state space.

Thus, a measurement in the orthonormal basis $\{|u_i\rangle\}_{i=1}^N$ corresponds to the set of projection operators $P_i = |u_i\rangle\langle u_i|$ and the probability of obtaining the base state $|u_i\rangle$ is $p_i = |\langle u_i|\psi\rangle|^2$. This postulate is called Born rule after Max Born, who proposed that the probability of measuring a particular eigenvalue (a possible outcome) of an observable (a measurable property) is proportional to the square of the absolute value of the corresponding coefficient in the wave function (a mathematical description of the quantum state) of the system (Messiah 2014). As a consequence of the Born rule, the *global phase* of a quantum state does not affect the statistics of measurement determined by the theory and can thus be ignored. That is, the states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$, where $\gamma \in \mathbb{R}$, are considered physically equivalent. To each dynamical state corresponds a vector defined to within a phase factor. Finally, it should be remarked that projective measurements together with unitary transformations are equivalent to general measurements.

Tensor product

A composite system is a system which cannot be described by a single particle. The state space of a composite system is the tensor product of the state spaces of the component systems. For a two-qubit system, the compact notation $|00\rangle = |0\rangle \otimes |0\rangle$ is used, where $\otimes$ denotes the tensor product. Note that $|00\rangle$ is a four-dimensional vector. The number of possible basis states increases exponentially with the number of qubits. A multi-qubit state is said to be *separable* if it can be written as a tensor product of single-qubit states. However, multi-qubit states are not necessarily separable. Non-separable states are called *entangled*. Manipulating a qubit in an entangled state can have effects on the other qubits. For example, the state

$$\alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle$$

is obviously separable. Separable states can be simulated efficiently on a classical computer, because the number of parameters needed to describe them is small. However, not all multi-qubit states are separable. States that cannot be written as a tensor product of individual qubit states are called *entangled* states. An example of an entangled state is the Bell state $|\Phi^+\rangle$:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (3)$$

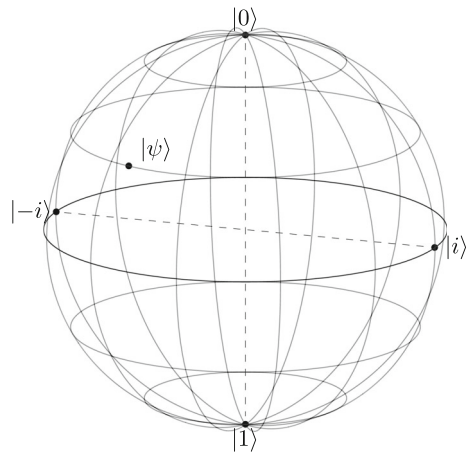
In a product state, the qubits are independent of each other, and measuring one qubit has no influence on the state of the other qubits. In other words, the outcome of measuring one qubit does not provide any information about the state of the remaining qubits. On the other hand, in an entangled state measuring one qubit can instantaneously affect the state of its entangled partners, even if they are physically separated. In maximally entangled states, measuring one qubit determines with certainty the outcome of measuring the other qubit. Entanglement is a fundamental resource in quantum computing, enabling the development of algorithms that can solve problems beyond the reach of classical computation. The unique properties of entangled states have profound implications and form the basis for the power of quantum computing. By harnessing entanglement, quantum algorithms can perform calculations and solve complex problems that are intractable for classical computers. The ability to manipulate and control the entangled states of multiple qubits is a crucial aspect of designing and implementing quantum algorithms, unlocking the potential for exponential speedup in certain computational tasks.

Pure states and mixed states

A superposition state can be known with certainty, even though the measurement outcome may be probabilistic. Superposition states are pointwise and they can in fact be visualized as points on the Bloch sphere $|\psi\rangle = \cos\frac{\theta}{2}|1\rangle + e^{i\phi}\sin\frac{\theta}{2}|0\rangle$. They are called *pure states*.

In contrast, if we are not sure if a qubit is in one pure state or another pure state, then the state itself is probabilistic, and not just the outcome of a measurement applied to the qubit. These states are called *mixed states*, and they can be visualized as a point in the Bloch ball. As an aside, the correspondence of this nomenclature, due to von Neumann Von Neumann (2018), with that of game theory, where classical strategies

Fig. 1 Bloch sphere



can be pure or mixed (Von Neumann and Morgenstern 1953) is no accident (Meyer 1999).

The formalism of density matrices allows to describe mixed states, defined above as probabilistic mixtures of kets with classical probabilities. In other words, the density matrix language provides a convenient means for describing quantum systems whose state is not completely known. While a pure state is written using vectors, a state that can be pure or mixed is written using a matrix called a density matrix. However common, the terminology is unfortunate, because a density matrix is an operator, not a matrix, and the matrix for this operator depends on the choice of orthonormal basis. In detail, suppose a quantum system is in one of a number of states $|\psi_i\rangle$ with respective probabilities p_i . $\{p_i, |\psi_i\rangle\}$ is an ensemble of pure states. The density matrix for the system is defined by the equation

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|. \quad (4)$$

Density operators must be positive (Hermitian and such that $\langle \psi | \rho | \psi \rangle \geq 0$ for any $|\psi\rangle$)—or, equivalently, have only non-negative eigenvalues), and have unit trace $\text{Tr } \rho = 1$. Actually, these properties can be used the other way around, to define density operators as operators that are positive and have unit trace. A density matrix ρ represents a mixed state if $\text{Tr } \rho^2 < 1$ and a pure state if and only if $\text{Tr } \rho^2 = 1$. In fact, the density matrix representing a pure state must be a projection and therefore be idempotent. The eigenvalues and eigenvectors of a density matrix have *no* special significance with regard to the ensemble of quantum states represented by that density matrix: they just indicate one of many possible ensembles that may give rise to that specific density matrix. This is related to the fact that the matrix representing a specific density operator depends on the basis.

Density operators can also be used to provide a partial description of a system. For example, given a composite quantum system consisting of two or more subsystems,

it can be meaningful to construct a description of just one of the subsystems, ignoring the others. Unitary operators act on density matrices by conjugation.

The von Neumann entropy is a measure of the uncertainty or mixedness of a quantum state. It is a generalization of the Shannon entropy quantifying the uncertainty of a classical probability distribution. The Von Neumann entropy of a density matrix ρ is

$$S(\rho) = -\text{Tr } \rho \log \rho$$

with logarithms being taken in base two. The von Neumann entropy is non-negative $S(\rho) \geq 0$, zero for pure states, and is maximized for the maximally mixed operator in a d -dimensional space: $S(I/d) = \log d$. The von Neumann entropy is in general smaller than the Shannon entropy for a mixture of states and they coincide if and only if the states in the mixture are orthogonal. This is associated with the indistinguishability on non-orthogonal quantum states.

We conclude this introductory account on Quantum Computing with a very interesting aspect called interaction-free measurement and an outline of the ways to realize qubits physically.

Interaction-free measurement

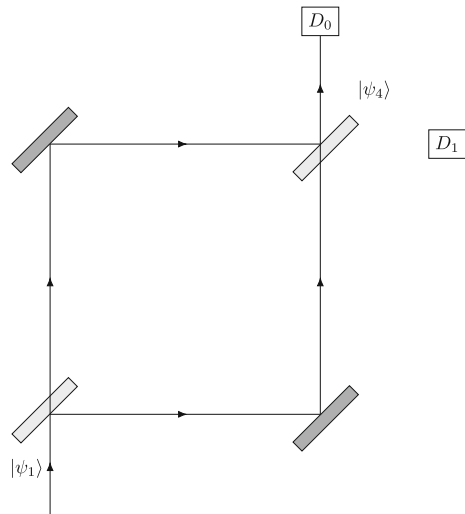
In a famous paper, Elitzur and Vaidman (1993) presented a gedankenexperiment based on a Mach-Zehnder interferometer where an obstacle can be put in one of the arms. The interferometer (Fig. 2) has a beam splitter, typically implemented with a half-silvered mirror, which divides the photon beam into two equal components. After reflecting off regular mirrors, the beams are recombined by another beam splitter, allowing for interference to occur. By adjusting the path length of each arm, it is possible to achieve constructive interference in one direction and destructive interference in the other, resulting in all photons exiting the interferometer in the same direction. Thus, the detector D_0 always fires, and D_1 never fires.

If an object is inserted into one of the arms (Fig. 3), interference is disrupted, and the photon can exit the interferometer in either direction, unless it is adsorbed by the object. Note that D_1 can only fire if the obstacle is present, even though it has not adsorbed the photon. The photon reaches D_1 in $\frac{1}{4}$ of the cases (half of the time it is adsorbed, and in the remaining $\frac{1}{4}$ it goes to D_2).

Physical quantum bits

In principle, any quantum system with two distinct states can be used as a qubit. Examples of systems that can be used as qubits: Photons, or particles of light, have a property called polarization that can be used as a qubit. Polarization states can be vertical, horizontal, or a superposition of these. Ions, which are atoms with an electric charge, can be trapped using electric fields. The two energy levels of a trapped ion can serve as a qubit. Atoms can be cooled to extremely low temperatures and trapped using magnetic fields and lasers. The nuclei of atoms and molecules have a property called spin, which can be used as a qubit. This spin can be detected by its precession frequency in a strong magnetic field. Quantum dots are tiny semiconductor devices that can trap an electron. The spin of this electron, which can be either “up” or “down,” can be used as a qubit. Defects in diamond crystals, such as a missing carbon atom

Fig. 2 The Mach-Zehnder interferometer in the Elitzur-Vaidman scheme



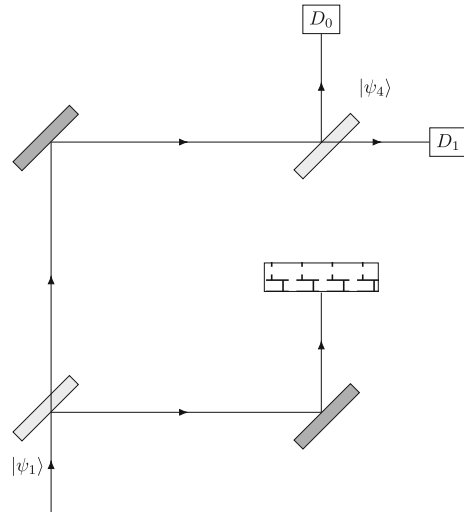
replaced with a nitrogen atom, can create a “spin triplet” that can be used for quantum computing. Superconducting circuits, which allow charge to flow with zero resistance, can be designed to have discrete energy levels. By adding a Josephson junction, these energy levels can be made distinguishable, allowing two of them to be used as a qubit.

The performance of quantum computing systems relies heavily on the characteristics of individual qubits, including their coherence times, operation speeds, and error rates (Hempel 2018). However, as the number of qubits increases, the complexity of hardware characterization and error analysis grows exponentially, making it challenging to maintain control and mitigate the effects of noise and errors. To achieve fault-tolerant operation at scale, researchers have developed the concept of logical qubits, which combine multiple physical qubits to enable quantum error correction through algorithmic redundancy. The efficiency of this approach is directly tied to the quality of the underlying physical qubits, highlighting the need for ongoing research into various qubit technologies. Despite significant advances, no single qubit modality has emerged as the clear winner, and the search for the most suitable qubit type continues.

3 Applications of quantum computing in finance

The application of Quantum Signal Processing (QSP) significantly reduces the quantum resources required for derivative pricing compared to previous approaches, including a 16x reduction in T-gates, 4x reduction in logical qubits, and 5x reduction in the required logical clock rate. Quantum advantage for derivative pricing will require 4.7k logical qubits and the ability to execute 10^9 T-gates at a rate of 45MHz (Stamatopoulos and Zeng 2024b).

Fig. 3 The interferometer of 2 with an obstacle in its right arm



3.1 Quantum option pricing

Pricing financial derivatives, in particular European-style options at different time-maturities and strikes, is a relevant problem in finance.

Quantum algorithms combining well-established numerical methods for stochastic differential equations and quantum amplitude estimation have been applied to the pricing of various options under stochastic volatility models, such as the Heston model (Wang and Kan 2023). It has been empirically found that the quantum algorithm based on weak Euler method achieves the same level of accuracy as the better-known strong Euler method, with a much higher efficiency due to eliminating the expensive procedure of preparing Gaussian states. The price of vanilla options with constant volatilities and interest rates is described by the Black-Scholes equation, a linear parabolic partial differential equation with terminal value given by the payoff of the option contract and no additional boundary conditions. If the Black-Scholes equation is made to correspond to the Schrödinger equation (2), the Hamiltonian is non-Hermitian. For anti-Hermitian Hamiltonians, the Schrödinger equation describes the dynamics of quantum states in imaginary time. While imaginary time evolution is non-unitary, it can be simulated on a quantum computer using the quantum imaginary time evolution (QITE) algorithm. The scope of application of QITE has also been expanded by lifting the restriction to anti-Hermitian Hamiltonians, thereby enabling the solution of any partial differential equation equivalent to the Schrödinger equation with an arbitrary, non-Hermitian Hamiltonian (Kumar and Wilmott 2024). Another approach to the non-Hermitian nature of the resulting Hamiltonian consists in embedding its propagator into an enlarged Hilbert space with an additional ancillary qubit (Gonzalez-Conde et al. 2023), with expected accuracy levels comparable to classical numerical algorithms by using nine qubits on a fault-tolerant quantum computer, and an expected success probability of the post-selection procedure due to the embedding protocol above 60%.

Recent advancements in solving high-dimensional partial differential equations (PDEs) have shown promising results in overcoming the curse of dimensionality. Quantum Monte Carlo algorithms have been developed for solving Black-Scholes PDEs in high dimensions (Li and Neufeld 2023), incorporating a wide range of payoff functions that are continuous and piecewise affine, which encompasses most relevant payoff functions used in finance. The computational complexity scales polynomially with the problem dimension and desired accuracy, demonstrating that it does not suffer from the curse of dimensionality.

For rainbow options, a specific type of path-independent multi-asset derivative, an end-to-end quantum circuit implementation using Iterative Quantum Amplitude Estimation has been developed, where the transition to price space is postponed for efficiency (Cibrario et al. 2024). Two amplitude loading methods are discussed for handling the exponential functions involved and experimental validation has been performed on the IBM QASM simulator.

Recent research has explored quantum computing applications for option pricing, particularly Bermudan options, where Miyamoto (2022) proposed a quantum algorithm quantum amplitude estimation and Chebyshev interpolation to approximate the continuation value, achieving a quadratic speed-up compared to classical Monte Carlo-based methods for Bermudan option pricing.

Scenario generation can be incorporated into quantum computations for financial risk analytics, by assembling quantum circuits that implement various stochastic risk factor models and integrate them with quantum amplitude estimation (QAE) algorithms (Matsakos and Nield 2024). The resource requirements for implementing financial risk analytics on quantum computers are estimated to be within the capabilities of near-term quantum computers in terms of the number of qubits ($\leq 1,200$), but the circuit depth ($\leq 10^8$) poses a significant challenge that needs to be addressed.

A fundamental step of any quantum algorithm is the preparation of qubit registers in a suitable initial state. Often qubit registers represent a discretization of continuous variables and the initial state is defined by a multivariate function. For preparing quantum states whose amplitudes encode multivariate functions (Rosenkranz et al. 2024) avoided the need to use arithmetic circuits, quantum Fourier transforms, or multivariate quantum signal processing and proposed developed protocols based on block-encodings of Fourier and Chebyshev basis functions. They also prepared bivariate Gaussian distributions on the Quantinuum H2-1 trapped-ion quantum processor using 24 qubits and up to 237 two-qubit gates.

Recent research has explored quantum speedups for stochastic optimization problems. Sidford and Zhang (2024) developed quantum algorithms for minimizing Lipschitz convex functions and computing critical points of smooth non-convex functions, achieving rates not known to be classically possible.

Systematic errors in the quantum algorithm can have a large and unpredictable impact, making consistent quantum advantage difficult to achieve with current quantum hardware (Udvarnoki et al. 2023). In detail, Udvarnoki et al. (2023) found that very low noise levels, with a two-qubit gate error rate below 10^{-6} , are necessary for the quantum method to outperform the classical one, and the Fourier series based approach for function approximation performs better than the traditional rescaling method.

Zhang et al. (2023) developed a photonic chip implementation of a quantum algorithm for pricing European options, leveraging the unary approach in conjunction with the quantum amplitude estimation algorithm. This innovative combination enables a quadratic speedup compared to classical Monte Carlo methods.

Variational quantum amplitude estimation (VQAE) is an emerging approach that combines variational quantum circuits with amplitude estimation techniques. Plekhanov et al. (2022) introduced VQAE, demonstrating its potential to outperform classical Monte Carlo sampling in certain scenarios. Variational quantum amplitude estimation uses shallow quantum circuits to approximate amplitude amplification and outperform classical Monte Carlo sampling in terms of computational cost, while keeping the quantum circuit depth below a fixed value. Adaptive VQAE, which rescales the amplitude to reduce the variational optimization cost, can further improve the performance of VQAE, which was not found to significantly depend on the number of qubits (up to 12 qubits tested).

Zhao et al. (2022) proposed a new adaptive algorithm for quantum amplitude estimation that uses Grover's algorithm and introduces an adjustment factor to improve the estimation process, achieving a quadratic speedup in the number of oracle queries compared to classical Monte Carlo sampling, but with a substantially lower computational complexity in the classical (non-quantum) part. The latest breakthrough by Labib et al. (2024) maps QAE to the classical direction of arrival estimation problem, allowing the use of signal processing algorithms for post-processing. Their approach achieves a total query complexity of $4.9/\epsilon$ and a parallel query complexity of $0.40/\epsilon$ at 95% confidence, surpassing previous best results. This connection between quantum and classical domains opens new avenues for QAE improvement by borrowing from the extensive literature in classical signal processing.

Recent research has explored quantum-enhanced Markov chain Monte Carlo (MCMC) methods, aiming to leverage quantum computing for faster sampling from complex probability distributions. Layden et al. (2023) demonstrated a quantum algorithm that performs MCMC to sample from the Boltzmann distribution of classical Ising models. The quantum algorithm converged in fewer iterations than common classical MCMC alternatives, suggesting unusual robustness to noise. Simulations showed a polynomial speedup between cubic and quartic over classical MCMC alternatives, which could ease computational bottlenecks in machine learning, statistical physics, and optimization.

Nakano et al. (2024) used a Quantum Alternating Operator Ansatz to enhance the convergence speed of Markov-chain Monte Carlo methods. They discovered a correlation between the acceptance rate of MCMC, the spectral gap of the MCMC transition matrix, and the convergence speed, which allows for optimizing the QAOA parameters to achieve a quadratic speedup in convergence. Recent research has explored quantum extensions of the Metropolis-Hastings (MH) algorithm, a widely used Markov chain Monte Carlo method. Miyamoto (2023) combined quantum Monte Carlo integration with quantum simulated annealing, obtaining improved scaling in cases where calculating the target distribution is costly. The log-likelihood is calculated by quantum Monte Carlo integration and quantum simulated annealing (QSA) generates the quantum state that encodes the target probability distribution in amplitudes. Quantum Dynamical Hamiltonian Monte Carlo (QD-HMC) (Lockwood et al.

2024) extends extend the well-known Hamiltonian Monte Carlo (HMC) method for MCMC, replacing the classical symplectic integration proposal step with simulations of quantum-coherent continuous-space dynamics on digital or analogue quantum computers.

3.2 Quantum portfolio optimization

The intersection of quantum computing and digital currencies has emerged as a vital area of research, bridging the fields of science, technology, and finance. One crucial yet underexplored aspect of this domain is the potential of quantum computing to enhance financial market trading and portfolio diversification. The study by Jabeur et al. (2024) investigates the dynamic relationships between quantum computing stocks and Bitcoin, revealing a strong time-scale dependence and demonstrating the performance of various portfolio strategies, including the Minimum Variance Portfolio (MVP), Minimum Correlation Portfolio (MCP), and Minimum Connectedness Portfolio (MCoP).

Quantum algorithms for portfolio optimization have shown promising results in recent years (Owolabi et al. 2024). Rebentrost and Lloyd (2024) proposed a quantum algorithm that can potentially achieve polynomial speedup over classical methods for determining the optimal risk-return curve. Venturelli and Kondratyev (2019) demonstrated that reverse quantum annealing, when seeded with a greedy local search solution, outperforms forward quantum annealing by a factor of 100 for hard instances, and a practical application has been recently proposed (Qu et al. 2024). Kerenidis et al. (2019) have also made notable progress in this area, developing the first quantum algorithm for constrained portfolio optimization, with experiments suggesting a potential linear speedup for most instances. Buonaiuto et al. (2023) focused on the best hyperparameters to perform portfolio optimization using the Variational Quantum Eigensolver (VQE) on real quantum computers, finding that the quality of the solutions obtained using VQE on real quantum devices strongly depends on the size of the quantum computer and the correct choice of hyperparameters. With the best hyperparameter choices, the quantum algorithm run on real quantum devices can reach solutions very close to the exact classical solution, with a strong convergence rate, even without error-mitigation techniques.

Quantum reinforcement learning methods using quantum neural networks can be applied to the problem of hedging in finance (Raj et al. 2023). Studies have compared classical and quantum algorithms through systematic backtesting, demonstrating that quantum approaches can match or slightly outperform classical results, particularly in handling larger asset numbers (Carrascal et al. 2023). Recent research demonstrates the potential of quantum machine learning (QML) to enhance financial forecasting (Mironowicz et al. 2024). Quantum-inspired classical methods and quantum neural networks (QNNs) have shown improvements in churn prediction and credit risk assessment, matching or surpassing classical models with fewer parameters (Thakkar et al. 2024). More recently, Kaushik et al. (2023) explored the use of Quantum Approximate Optimization Algorithm (QAOA) and Variational Quantum Eigensolver (VQE) to maximize the Sharpe ratio, comparing the results of both approaches. These studies

highlight the potential of quantum computing to provide more efficient and accurate solutions for complex portfolio optimization problems.

3.3 Quantum machine learning

Quantum Machine Learning (QML) has found many applications in quantitative finance. The intersection of quantum computing and machine learning in financial contexts touches an diverse array of use cases, such as risk management, credit assessment, portfolio optimization, and derivatives pricing (Mironowicz et al. 2024).

The complexities and uncertainties of financial markets pose significant challenges for predicting market trends and developing optimal trading strategies, and quantum-classical hybrid algorithms which combine the strengths of quantum computing and reinforcement learning are one of the methods that have been applied to address these challenges. Dutta et al. (2024) introduced the Quantum Attention Deep Q-Network (QADQN), where variational quantum circuit is used with a traditional deep Q-learning framework. Backtesting using historical data from major market indices, including the S&P 500, and realistic transaction costs. Our results show that the QADQN agent outperforms traditional methods.

Quantum computing shows promise for enhancing financial risk measurement, potentially offering significant speed-ups over traditional methods, improving risk assessment strategies (Adegbola et al. 2024). Conceptual solutions and small-scale quantum circuits for financial risk measurement are feasible, but the hardware capacity and quantum noise control needed for real-world deployment are still significant challenges, because about 100 qubits are still largely insufficient to build a usable risk measurement system (Wilkins and Moorhouse 2023). In addition, quantum noise poses an additional challenge that needs to be addressed through further research before quantum-based risk measurement applications can be deployed in practice.

Quantum machine learning (QML) is an emerging field that combines quantum computing and classical machine learning to potentially accelerate and enhance various computational tasks (Sarkar 2023; Mishra et al. 2021). QML explores quantum-enhanced algorithms, hybrid quantum-classical neural networks, and the application of machine learning to quantum hardware (Melnikov et al. 2023). The field aims to leverage quantum properties to improve pattern recognition, data processing, and predictive capabilities beyond classical limits (Mishra et al. 2021). Potential applications include drug discovery, speech and image recognition, and financial modeling (Sarkar 2023). However, QML faces challenges in practical quantum computing resources, algorithm design, and error mitigation (Sarkar 2023; Melnikov et al. 2023). Despite these obstacles, QML shows promise in various domains, including physics, chemistry, and natural language processing. Continued interdisciplinary collaboration and advancements in quantum hardware and software are crucial for realizing the full potential of QML (Sarkar 2023; Melnikov et al. 2023).

4 Quantum theory: an application to decision theory

4.1 Surprising connections

Recent studies have shown that quantum structures, such as contextuality, emergence, entanglement, indistinguishability, interference, and superposition, are not exclusive to the realm of physics. Instead, these structures have been observed in various other domains, including cognition, where they influence human probability and similarity judgments, decision-making, language, and perception. Similarly, structures similar to the quantum ones have been found in socio-economic domains, such as behavioral economics and finance, where empirical data often deviates from the predictions of classical logical and probabilistic frameworks. In the field of information systems, including computer science and artificial intelligence, quantum structures have also been identified. This phenomenon has significant implications for our understanding of complex systems and the development of new computational models. The presence of quantum structures in these domains challenges the traditional Boolean and Kolmogorovian approaches, which have been the foundation of classical logic and probability theory. By examining the role of quantum-like structures in these domains, we hope to deepen our understanding of the underlying principles and mechanisms that govern complex systems, ultimately leading to the development of innovative computational models and applications.

While there is no evidence supporting the idea that the brain is a quantum system, it is nevertheless useful to use the nominal analogy between decision making with emotions and quantum measurement with noise to simplify the descriptions and make better predictions to be confirmed or disproved by experiments (Yukalov 2023). In particular, quantum interference, that is the violation of the law of total probability which occurs in quantum mechanics when two observables are not compatible (they do not commute) can account for a variety of problematic situations in decision theory (Pothos and Busemeyer 2022). One of the most significant rational decision theories is von Neumann and Morgenstern's expected utility (EU) theory (Von Neumann and Morgenstern 1953). Expected utility theory is intended for decisions under risk, meaning choices among actions defined by an objective probability distribution over possible payoffs. An important extension of EU theory is Savage's subjective expected utility (SEU) theory (Savage 1972). This extends utility theory to decisions under uncertainty, where choices among actions are defined as functions mapping uncertain events to payoffs, and these events may not have objective probabilities. The widespread adoption of SEU can be attributed to its intuitive and mathematically rigorous foundation, as formalized by Savage. Nevertheless, its enduring influence, particularly in statistical decision-making and learning theory, is also significantly due to De Finetti (1937). Human behavior often deviates from the rational axioms of Expected Utility (EU) theory and Subjective Expected Utility (SEU) theory. As far as the generalizations of Decision researchers have created specific types of decision-making scenarios involving risk, where individuals consistently violate EU axioms, as well as scenarios involving uncertainty, where individuals regularly violate SEU axioms (Busemeyer and Bruza 2012). These two instances of behavioral deviations

Table 1 Ellsberg option sets

Action	R	B	G
<i>First set of options</i>			
<i>A</i>	\$100	0	0
<i>B</i>	0	\$100	0
<i>Second set of options</i>			
<i>A'</i>	\$100	0	\$100
<i>B'</i>	0	\$100	\$100

from rational decision-making are known as the Allais and Ellsberg paradoxes, respectively. As far as the generalizations of utility theory and related paradoxes, we refer to (Coletti G. 2002), (Schmeidler 1989).

In the following we will refer to the Ellsberg’s 1961 classic, taking into account the considerations of Busemeyer and Bruza (2012) and Moscati (2024).

4.2 The Ellsberg paradox

In 2009, La Mura introduced an axiomatic framework for a quantum theory of decision-making under uncertainty. This theory was employed to explain the Allais and Ellsberg paradoxes. Here, after a description of the classic Ellsberg paradox, with additive and non additive decision weights, we provide a concise overview of the theory and its application to the paradox.

4.2.1 The Ellsberg paradox with additive decision weights

The decision maker (player) knows the following facts:

- 1. a ball will be randomly drawn from an urn.
- 2. The urn contains red, blue, and green balls.
- 3. The urn holds 100 red balls and 200 additional balls, but the exact number of blue and green balls is unknown.

Consider the first set of options between actions *A* and *B* in Table 1. If action *A* is selected and a red ball is drawn, the player wins \$100; otherwise, he doesn’t win anything. If action *B* is selected and a blue ball is drawn, the player wins \$100; otherwise he doesn’t win anything. Now, consider the second set of options. If action *A'* is chosen, the player wins \$100 if either a red or green ball is drawn; otherwise, he doesn’t get anything. If action *B'* is chosen, the player wins \$100 if either a blue or green ball is drawn; otherwise, he doesn’t get anything. The first and the second set of options, already described, have been created to test the *sure-thing* principle due to Savage (1972). In decision theory, the *sure-thing* principle states that a decision maker who decides to take a certain action in the case that an event *E* occurs, as well as in the case that the negation of *E* ($\bar{E}$) occurs, should also take that same action if he knows nothing about *E*.

Let us consider the first set of options in Table 1:

- if $\bar{G}$ occurs, it is not so evident which is the option preferred, let us suppose, at the moment, that $A \succeq B$.
- If G occurs, A and B give the same payoff thus it is $A \succeq B$.

Therefore, for the *sure-thing* principle, $A \succeq B$ even if G is unknown.

Let us consider the second set of options in table 1:

- if $\bar{G}$ occurs, then A' is A and B' is B , so $A' \succeq B'$.
- If G occurs, A' and B' give the same payoff thus it is $A' \succeq B'$.

Therefore, also in this case, for the *sure-thing* principle, $A' \succeq B'$ even if G is unknown.

Now, let us change the original assumption for the first pair of options to $B \succeq A$ when $\bar{G}$ occurs. Then, following the same reasoning as before, it follows that $B' \succeq A'$.

Summarizing, for the *sure-thing* principle:

$$A \succeq B \Rightarrow A' \succeq B' \text{ or } B \succeq A \Rightarrow B' \succeq A'.$$

Let us indicate with $P(E_i)$ the subjective probability that a decision maker assigns to the occurrence of event E_i , and let us indicate with $u(act|E_i)$ the utility of the payoff obtained by an action act when event E_i occurs. According to SEU theory the decision maker decides to choose the action that maximizes the *subjective expected utility*:

$$SEU(act) = \sum_i p(E_i) \cdot u(act|E_i).$$

Referring to the Ellsberg choice set in Table 1, it is:

$$\begin{aligned} SEU(A) &= p(R) \cdot u(\$100), \\ SEU(B) &= p(B) \cdot u(\$100), \\ SEU(A') &= p(R) \cdot u(\$100) + p(G) \cdot u(\$100), \\ SEU(B') &= p(B) \cdot u(\$100) + p(G) \cdot u(\$100), \end{aligned}$$

thus, $A \succ B \Rightarrow p(R) > p(B) \Rightarrow A' \succ B'$. The shared outcome resulting from event G is *neutralized* and has no impact, making the decision between A and B equivalent to the decision between A' and B' .

In summary:

$$p(R) \geq p(B) \Leftrightarrow A \succeq B \Leftrightarrow A' \succ B',$$

but

$$p(B) \geq p(R) \Leftrightarrow B \succeq A \Leftrightarrow B' \succ A'.$$

The Ellsberg paradox highlights that empirical studies on the Ellsberg problems in Table 1 consistently show that, for the first pair of choices, people predominantly select option A over B . However, for the second pair of choices, the same individuals tend to prefer option B' over A' (Camerer and Weber 1992).

4.2.2 The Ellsberg paradox with non additive decision weights

To account the Ellsberg paradox, according to Tversky and Fox (1995), prospect theory with nonadditive decision weights may be used as described here below. For A and B the *prospect weighted utility* are respectively:

$$\begin{aligned}PWU(A) &= w(R) \cdot u(\$100), \\ PWU(B) &= w(B) \cdot u(\$100),\end{aligned}$$

thus, $A \succ B \Rightarrow w(R) > w(B)$. For the second pair of actions, instead, it is:

$$\begin{aligned}PWU(A') &= w(R \cup G) \cdot u(\$100), \\ PWU(B') &= w(B \cup G) \cdot u(\$100),\end{aligned}$$

thus, $B' \succ A' \Rightarrow w(B \cup G) > w(R \cup G)$, which requires that w is non-additive for mutually exclusive events that is, $w(X \cup Y) \neq w(X) + w(Y)$ for $X \cap Y = \emptyset$.

Most researchers consider the findings related to the Ellsberg paradoxes as evidence that the principles of expected utility theory are systematically violated. However it is important to highlight that principles of expected utility theory pertain to the *preferences* of individual decision-makers, whereas the empirical results are derived from *choices averaged* across multiple individuals. The evidence implies a violation of the expected utility theory only if we assume homogeneity across individuals but this is not a reasonable assumption. Moreover, one might contend that an expected utility principle is violated whenever an individual exhibits a choice pattern of selecting act A in the first pair and then act B' in the second pair. However, it is important to remark that choice is probabilistic, thus a single choice pattern by an individual is an unreliable indicator of underlying preference relations (Luce et al. 2000). Indeed, Camerer et al. (1989) discovered that in his experimental tests of utility models, participants changed their choices in almost one out of every three trials, even when actual payoffs were at stake. Given the probabilistic nature of choice, testing utility principles necessitates estimating the probability of choice patterns for each individual. To convincingly argue that an axiom is violated, greater attention to individual differences and statistical methodology is essential (Regenwetter et al. 2010).

4.3 Quantum theory for the Ellsberg paradox

Let us consider a set $\Omega = \{\omega_1, \dots, \omega_n\}$ of n states of the world and a probability measure $P = \{p_1, \dots, p_n\}$ defined on it. Suppose that $X = \{x_1, \dots, x_n\}$ is a risky action that produces the real valued payoff x_i when the state ω_i occurs with probability p_i . Let us consider the $n \times 1$ vector $\Psi = [\sqrt{p_1}, \dots, \sqrt{p_n}]'$ so that $\Psi' \cdot \Psi = 1$. The risky action is evaluated by an $n \times n$ Hermitian matrix denoted by U ; the i^{th} eigenvalue of U , $u(x_i)$, equals the utility of payoff x_i and V_i is the corresponding eigenvector. The *projective expected utility* (PEU) for the action X , corresponding to the amplitude

matrix Ψ , is:

$$\begin{aligned}
 PEU(\Psi) &= \Psi' \cdot U \cdot \Psi = \\
 &= \Psi' \left(\sum_i u(x_i) \cdot V_i \cdot V_i' \right) \Psi \\
 &= \sum_i u(x_i) \cdot (\Psi' \cdot V_i) \cdot (V_i' \cdot \Psi) \\
 &= \sum_i u(x_i) \cdot |\Psi' \cdot V_i|^2.
 \end{aligned} \tag{5}$$

Thus, the projective expected utility for the risky action X is a weighted sum of the utilities of the outcomes x_i each of which with weight given by $|\Psi' \cdot V_i|^2$. It has been observed that the act with the highest value of PEU is more often chosen.¹

Let us now apply this theory to the Ellsberg paradox: there are only two payoffs (\$100, 0) and the corresponding 2×2 Hermitian utility matrix U

$$U = \begin{pmatrix} 1 & t \\ t & 0 \end{pmatrix}$$

where t is a real free parameter. The quantity of red balls is certain, but the number of blue and green balls is unknown. Consequently, the decision maker must evaluate different potential distributions of blue and green balls. The player faces three possible different scenarios:

1. all balls are blue;
2. all balls are green.
3. the number of blue balls equals the number of green balls;

Let us consider the first pair of actions A and B . Choosing action A , since the number of red balls is known, it is also known that the probability to win is exactly $1/3$ thus:

$$\begin{aligned}
 PEU(A) &= \left[\sqrt{1/3} \sqrt{2/3} \right] \cdot \begin{pmatrix} 1 & t \\ t & 0 \end{pmatrix} \cdot \begin{bmatrix} \sqrt{1/3} \\ \sqrt{2/3} \end{bmatrix} = \\
 &= \frac{1}{3} + \frac{\sqrt{8}}{3} \cdot t = \frac{1}{3} + \frac{\sqrt{72}}{9} \cdot t.
 \end{aligned}$$

Choosing action B , instead, the payoffs depend on the number of blue balls which is unknown, thus, a mixture across possible states is necessary in this case:

$$PEU(B) = \frac{1}{3} \cdot \left(\frac{2}{3} + \frac{\sqrt{8}}{3} \cdot t \right) + \frac{1}{3} \cdot \left(\frac{1}{3} + \frac{\sqrt{8}}{3} \cdot t \right) + \frac{1}{3} \cdot 0 =$$

¹ Let us notice that, if U is a diagonal matrix, PEU reduces to the von Neumann and Morgenstern EU theory.

$$= \frac{1}{3} + 2 \cdot \frac{\sqrt{8}}{9} \cdot t = \frac{1}{3} + \frac{\sqrt{32}}{9} \cdot t$$

It follows that:

$$t > 0 \Leftrightarrow PEU(A) > PEU(B)$$

Now, let us take into account the second pair of actions A' , B' . Choosing option B' , the payoffs depend on the number of blu and green balls which is known, thus the probability to win is $2/3$ without uncertainty and we have:

$$\begin{aligned} PEU(B') &= \left[\sqrt{2/3} \sqrt{1/3} \right] \cdot \begin{pmatrix} 1 & t \\ t & 0 \end{pmatrix} \cdot \left[\begin{matrix} \sqrt{2/3} \\ \sqrt{1/3} \end{matrix} \right] = \\ &= \frac{2}{3} + \frac{\sqrt{8}}{3} \cdot t = \frac{6}{9} + \frac{\sqrt{72}}{3} \cdot t. \end{aligned}$$

Choosing action A' , instead, the payoffs depend on the number of green balls which is unknown, thus, a mixture across possible states is necessary:

$$\begin{aligned} PEU(A') &= \frac{1}{3} \cdot \left(\frac{1}{3} + \frac{\sqrt{8}}{3} \cdot t \right) + \frac{1}{3} \cdot \left(\frac{2}{3} + \frac{\sqrt{8}}{3} \cdot t \right) + \frac{1}{3} \cdot 1 = \\ &= \frac{6}{9} + 2 \cdot \frac{\sqrt{8}}{9} \cdot t = \frac{6}{9} + \frac{\sqrt{32}}{9} \cdot t \end{aligned}$$

It follows that:

$$t > 0 \Leftrightarrow PEU(B') > PEU(A')$$

As a result, the key factor in deciding whether to approach or avoid uncertainty is the off-diagonal element of the utility matrix:

- if $t > 0$ risk is preferred to uncertainty;
- if $t = 0$ we are in the case of expected utility theory where the decision maker is indifferent between risk and uncertainty.
- if $t < 0$ decision maker prefers uncertainty to risk.

La Mura (2009) also illustrates how the PEU model can explain the Allais paradox, though we direct readers to the original article for further details. In summary, quantum decision-making models can address both the Allais and Ellsberg paradoxes, as can non-additive weighted utility models.

4.4 Theoretical results

Quantum theoretical finance applies quantum theory principles to financial modeling, offering new perspectives on option pricing and market dynamics (Dong et al. 2024). Interestingly, the formal framework of quantum mechanics arises in economics

generally and in financial economics in particular as a natural consequence of an information-theoretic approach to these fields (Hawkins and Frieden 2017). Specifically, extremizing Fisher information, subject to the constraint that the associated probability density reproduces observed economic phenomena, results in equations similar to the Schrödinger equation (2). Established finance models can also be re-interpreted through the lens of quantum physics. For example, drawing inspiration from the well-known phenomenon of quantum tunneling, where particles defy classical expectations by traversing energy barriers, in the quantum financial leapfrog model (Zheng and Bai 2024), asset prices exhibit quantum-like behavior, making non-sequential jumps that penetrate market “energy barriers”. Nastasiuk (2014) proposed a theory of quantum-like dynamics in financial markets, with a non-differentiability hypothesis for the price-time continuum and fractal properties. Quantum finance models replace random variables with operators and account for market self-reflexivity and incompatible observations (Focardi et al. 2020). Chakrabarti et al. (2021) introduced a new re-parameterization method for quantum derivative pricing that significantly reduces the resource requirements compared to previous approaches. Chakrabarti et al. (2021) also provided complete resource estimates for quantum derivative pricing, finding that pricing autocallable and TARF derivatives would require 8k logical qubits and a T-depth of 54 million, which would need to be executed in around 1 s to achieve quantum advantage.

5 Limitations of quantum technologies in finance and potential breakthroughs

Quantum technologies are poised to revolutionize the field of finance, yet several limitations and challenges remain. Currently, quantum computers are in an experimental stage, featuring immature and error-prone hardware that restricts their practical applications. The scalability of qubits, the fundamental units of quantum information, presents a significant challenge, as current systems cannot handle enough qubits for complex financial tasks. Additionally, quantum computers are highly sensitive to noise and errors, and while quantum error correction is an active area of research, it has not yet been fully resolved. Integrating these technologies with traditional financial infrastructures necessitates substantial collaboration between financial institutions, technology providers, and regulatory authorities. Moreover, the development and maintenance of quantum computers are extremely costly, limiting their accessibility for many financial institutions.

Despite these hurdles, potential breakthroughs and future advancements could transform the field. A substantial increase in the number of physical and logical qubits with low error rates could make quantum computers more viable for complex financial applications. The development of new quantum algorithms could enhance portfolio optimization, risk modeling, and other financial applications. Advances in quantum cryptography could secure financial transactions against future threats. The integration of artificial intelligence with quantum computing could result in more accurate predictions and personalized investment strategies. Collaborations between financial

institutions and technology companies, along with government investments, could accelerate the development and adoption of quantum technologies.

Despite the challenges, the potential of quantum technologies in finance is vast. With ongoing progress, we could witness a significant transformation in how financial data is managed and investment decisions are made.

6 Challenges

The main uncertainty surrounding quantum computers in the coming years is whether any applications can be achieved using current or near-future quantum computers without, or with only minimal, error correction.

Both error-corrected and non-error-corrected quantum computers are subject to noise, but they differ in the degree of noise they produce. In general, error-corrected quantum computers, assuming they have sufficiently low physical error rates, can be made exponentially less noisy than their non-error-corrected counterparts by utilizing error-correcting codes. The viability of a specific use case or application largely depends on how much noise can be tolerated in the execution of the required circuits before the output is compromised.

The smaller the circuit size required, the more likely it is that the circuit can be successfully executed on current or near-future quantum computers. This increases the potential to realize use cases or applications that depend on such circuits. Therefore, the question of whether near-future quantum computers can run useful applications is closely linked to whether small-sized circuits can achieve this.

Three major trends support this possibility: the development of new quantum algorithms that require small or shallow-depth circuits, innovative techniques for managing noise in quantum circuit execution, and methods for breaking down large circuits into smaller ones. These trends all involve repeated execution of diverse circuits and are compatible with current and near-future quantum computers. While achieving quantum error correction remains crucial for realizing the full potential of quantum computing, it may still be possible to address interesting problems with near-future quantum computers, even without full fault tolerance.

The first trend involves the development of new quantum algorithms, particularly variational algorithms that can be customized for both the problem and the specific quantum computer. This flexibility allows the scaling up of circuit size to the limits of the hardware's capabilities, though it requires running numerous circuits. These algorithms can be executed on currently available quantum computers, making it easier for them to run sufficiently complex circuits for scientific or commercial purposes.

The second trend focuses on new techniques for managing noise, specifically quantum error mitigation. These techniques involve repeatedly executing a set of circuits and using classical post-processing to estimate the output that would be obtained from a noiseless quantum computer. Essentially, they extract more signal from noisy quantum hardware. Error mitigation can be applied to both variational and non-variational algorithms, extending the hardware's capability to run larger circuits.

The third trend, known as circuit knitting, involves decomposing large circuits into smaller ones. This family of techniques reduces circuit size (or depth) by decreasing

the number of qubits (or timesteps) required, at the cost of needing to run more circuits. The results of a larger circuit are computed through classical post-processing of the outputs from smaller circuits. Because these smaller circuits can be run on current and near-future quantum computers, they incur less error than attempting to run the larger circuit as a whole.

7 Conclusion

The unique properties of quantum computing, including superposition, entanglement, parallelism, and interference, are the key drivers for the potential of quantum computers. By leveraging superposition and entanglement, qubits can be manipulated to efficiently encode and process vast amounts of information in parallel, enabling the exploration of an exponentially large solution space. Furthermore, quantum interference plays a crucial role in eliminating unwanted by-products of parallel computation and amplifying the desired results, facilitating efficient readout of the correct solutions. Despite the challenges in developing quantum algorithms that can harness the power of parallel computation while providing an efficient means of extracting results, significant progress has been made in recent years. The list of quantum algorithms capable of achieving this feat is growing, holding great promise for the future of quantum computing.

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